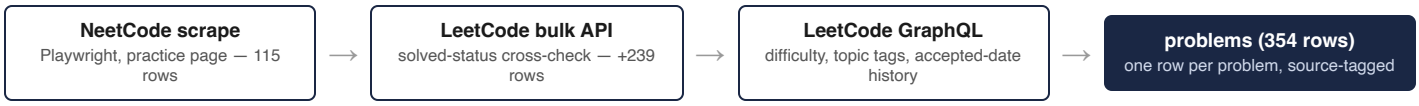
Problem Tracker — Segment 6: Two Data Sources, One Schema

6 of 8 · 5 min

Reconciling NeetCode + LeetCode · your specific involvement in a hard, ongoing technical problem

Opening line: "NeetCode only knows about problems solved through its own editor — solve the same problem directly on leetcode.com and NeetCode never finds out. The real source of truth is two systems that don't talk to each other, and I had to reconcile them into one identity per problem."



The 5-Minute Run of Show

0:00–0:20	Opening line (above)
0:20–1:00	The identity problem + the fix
1:00–1:45	The sync pipeline, three passes
1:45–2:25	Data-safety guarantees
2:25–4:00	Extending the reconciliation live: difficulty + category
4:00–5:00	Why one schema, not one table per source + close

Numbers to Have Cold

- **354** total problems (239 LeetCode / 115 NeetCode)
- **1** discriminator column, not 2 tables — `source IN ('neetcode','leetcode')`
- **62**-entry slug alias table, grown from 17 by diffing real gaps

Section-by-Section Script

The identity problem + the fix (0:20–1:00)

- *"NeetCode sometimes hosts the same problem under a different URL slug than LeetCode's canonical one. Without handling that, syncing LeetCode's solved list would insert a duplicate row for a problem I'd already tracked from the NeetCode side."*
- *"Fix: a slug-alias map, built both directions, normalizing every sync step before it touches the DB."*

The sync pipeline, three passes (1:00–1:45)

- *"tracker.py scrapes NeetCode for what I've completed there, cross-checks LeetCode's own bulk solved-problems API for anything solved outside NeetCode, then backfills difficulty and topic tags for every row from that same LeetCode endpoint."*

Data-safety guarantees (1:45–2:25)

- *"The sync never overwrites an existing row's category or source — it only bumps last_seen. A genuinely new discovery gets inserted as source='leetcode', never guessed into an existing category. Per-problem fetches are capped per run so one run can't fire off hundreds of extra requests."*

Extending it live: difficulty + category (2:25–4:00)

- *"Difficulty came almost free — LeetCode's bulk endpoint already returns it for every problem it knows about, solved or not. Category was harder: LeetCode gives multiple topic tags per problem, not one, so I had to pick a single 'primary' tag, deprioritizing overly generic ones like 'Array' in favor of something more specific, and translate it onto an existing NeetCode bucket name where there's an obvious match."*
- *"This same reconciliation logic is also exactly where a real bug showed up recently — a URL-identity issue that created 127 duplicate rows. More on that in the mistake-story segment."*

If they ask "what happens if the two sources disagree"? — they can't really disagree on the same fact, only on coverage: NeetCode's checkbox reflects solves through its own editor; LeetCode's API reflects an accepted submission anywhere. A problem solved outside NeetCode legitimately shows unsolved there — that's the gap the cross-check exists to fill, not a conflict to resolve.